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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Case Study **Duration: 2hrs**

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. **(5 Marks)**

2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**

4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**

5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**

6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**

7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance.**(5 Marks)**

8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. **(5 Marks)**

9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. **(5 Marks)**

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

I S·Roshan (192425258) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

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Develop a python based Reinforcement learning model using open AI Gym to gold world navigation

1) **Problem understanding:**

The objective is to develops a (RL) agent that navigates through a grid world environment.

2) **Flow Chart:**

Start
↓
create grid world
↓
Initialize Q-Table
↓
choose Action (ε-greedy policy)
↓
move to Nearest state
↓
Receive reward
↓
updates Q-Table
↓
goal reached?
↓ ↓
NO Yes

3) **Reinforcment learning components**

components	Description
enivronment	5x5 grid world
Agent	RL Agent
state space	each grid cell represents one state
Action space	UP, Down, Left, right
goal state	Reach destination safety
obstacle state	

4) Reward function

situation	rewards
Reach goal	+100
hit abstacle	-50
normal move	-1
Invalid move	-10

5) constraints

constraint	purpose
Avoid obstacles	collision
stay withing goals	valid
Reach goal	navigation
minimize steps	efficiency

6) Python Implementation:

```
Impact gym
Import numpy as np
Q = np.zeros ((25,4))
alpha = 0.1
gamma = 0.9
epsilon = 0.1
```

7) Working Process:

1) create the grid word environment
2) Intialize the Q-table with zero values.

5) Receive reward from agent reaches the goal

6) Store the learned policy for navigation

8) **Belman equation:**

$$Q(s,a) = Q(s,a) + \alpha(R + \gamma \max Q(s',a') - Q(s,a))$$

9) **Conclusion:**

The Q-learning based (RL) model successfully learns an optimal navigation policy.

2) **Greedy multi-formed Bandit Algorithm:**

Problem statement:

The e-greedy algorithm helps an agent maximize rewards heps on agent maximize rewards by babrading exporation within 500 tasks

Flow chart:

Start

↓

Initialize Args

↓

Choose value

↓

Select Arm
↓
Get reward
↓
Update reward estimate
↓
500 trials
↓
No → Repeate
Yes → compose results

<u>Parameters:</u>

Parameters	value
Arms	5
Traits	500
evalues	0.1, 0.3, 0.5

<u>comparision:</u>

E	exptation	Performance
0.1	low	Best reward
0.3	reward	Balanced
0.5	high	lower reward

<u>conclusion:</u>
The e=0.1 policy provide the highest circuative rewards by balancing Amitted exeploration

with maximize exploratation under the 500 trail constraint.

3) Design a Markov Decision Process (MDP) for an Autonomus Robot:

Problem statement:

Design an MDP model for an autonomous robot that reaches its destination while reducing energy usage and avoiding unnecessary movement

Flow chart:

Start
↓
Identity current state
↓
select Best Action
↓
Move to nearest state
↓
update rewards & energy
↓
Destination reached?
↳ No → continue
↳ yes → stop.

Python Syntax:

state = ["Start", "Path", "Goal"]

actions = ["up", "Down", "left", "Right"]

rewards = 100

battery - limit = 30.

Conclusion:

The MDP enables the robot to choose the optimal path while making energy consumption Improving navigation efficiency and battery utilization

4) **Value iteration Algorithm for a constrained grid environment**

Problem statement:

implement a value iteration algorithm using python to find the optional policy in a grid environment with restorted states

flow chart:

Start
↓
Initial State values
↓
Apply Bellaman equation.

Update state values

↓

Check convergence

↓ No → Repeat

↓ Yes → optimal policy

Bellman Equation

$$V(S) = \max_a [R + \gamma V(S')]$$

Python syntax:

```
Impart tensorflow as tf
Imaport numpy as np
Jamma=0.9
values= np.zeros((5,5))
```

Conclusion:

The value Iteration algorithm applies the Bellaman equation to find the optimal poliry while avoiding restricted states and maraiming long term rewards.

5) **RL Framework for a warehouse robot with Battery constant**

Problem statement:

Design a (RL) frame work for a warehouse robot that completes the maximum no at deserveres without execeeding a 30-minute Battery Limit

Flowchart:

Start
↓
Check battery level
↓
select Delivers Task
↓
move to Destination
↓
Delivery Package
↓
updates Battery a reward
↓
Battery Available?
↳ yes → continue
↳ NO → stop

Python syntax:

```
battery = 30
reward = 50
If battery > 0;
    action = 'Delivers'
```

Conclusion!

The RL frame work helps the warehouse robot maxizine deseverries while staying within the battery.